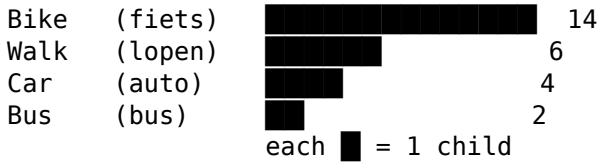
7.8 Practice: graphs and averages

← Unit 7.8 · Answers

Graph routine: **1.** read the title, **2.** look at what's on each axis, **3.** check the scale, **4.** then answer the question.

Set 1 · Warm-up: bar chart (*staafdiagram*)

How do the children in groep 7B get to school?



1. How many children come by bike?
2. How many children walk?
3. Which way of getting to school is the most common?
4. How many more children cycle than walk?
5. How many children are in the class altogether?
6. How many children come by car or bus?
7. How many children do *not* come by bike?
8. Which bar is the shortest?

Talk about it: Is this class probably in a city or in the countryside? What makes you think so? Can you be sure?

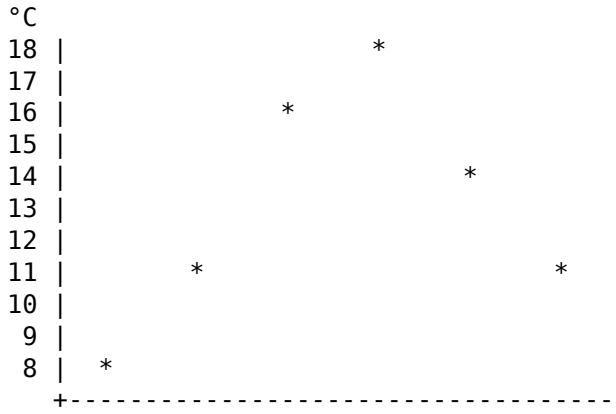
Set 2 · Building: line graph (*lijngrafiek*)

Temperature in the garden on 1 May

Time	06:00	09:00	12:00	15:00	18:00	21:00
Temperature (°C)	8	11	16	18	14	11

First draw this as a line graph on squared paper: time along the bottom, temperature up the side.

Here are the points (join them with straight lines on your own graph):



```
  06    09    12    15    18    21   time
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1. At what time was it warmest?
2. How many degrees did the temperature rise between 06:00 and 12:00?
3. How many degrees did it fall between 15:00 and 21:00?
4. Between which two times did the temperature rise the most?
5. What is the difference between the warmest and the coldest temperature?
6. What is the average of the six temperatures?
7. At which two times was the temperature the same?
8. What was the temperature at 12:00?

Talk about it: Why is a line graph a good choice here, and a pie chart a bad one?

Set 3 · Building: average (*gemiddelde*)

Work out the average.

1. 4 and 6
2. 3, 5 and 7
3. 10, 20, 30 and 40
4. 2, 4, 6, 8 and 10
5. Four friends get pocket money (*zakgeld*): € 5, € 7, € 4 and € 8 a week. What is the average?
6. Ilse cycles to school and back. From Monday to Friday she cycles 3, 3, 5, 3 and 6 km. What is her average per day?
7. Your test marks are 7, 8, 6, 9 and 5. What is your average mark?
8. Three sunflowers are 1,5 m, 1,8 m and 2,1 m tall. What is their average height?
9. The average of four numbers is 10. Three of the numbers are 8, 12 and 9. What is the fourth number?
10. 5, 5 and 5

Talk about it: In question 6, Ilse cycled 4 km on *no* day. So how can 4 km be her average?

Set 4 · Building: pie chart (*cirkeldiagram*)

Favourite fruit of 40 children

The pie chart is split like this:

- Apple: half of the circle
- Banana: a quarter of the circle
- Strawberry: an eighth of the circle
- Mango: an eighth of the circle

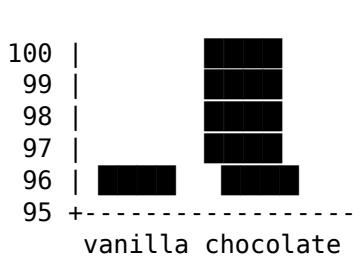
(Draw it: a circle, split in half, then one half split into a quarter and two eighths.)

1. How many children chose apple?
2. How many children chose banana?
3. How many children chose strawberry?
4. What percentage of the children chose banana?
5. Another school asks 80 children and gets a pie chart with exactly the same shape. How many children there chose apple?
6. 10 out of 40 children chose banana. What fraction is that, as simple as possible?
7. How many children chose strawberry *or* mango? What fraction of the circle is that?
8. Which fruit takes up half of the circle?

Talk about it: Could you draw this information as a bar chart too? Which do you find easier to read?

Set 5 · Stretch: thinking critically (*kritisch kijken*)

A. The ice-cream graph. An ice-cream shop sold 96 vanilla cones and 100 chocolate cones. Its advert shows this graph:



1. How many more chocolate cones than vanilla cones were sold?
2. On the graph, the chocolate bar looks how many times as tall as the vanilla bar?
3. Was chocolate really that many times as popular? What trick is the graph using?

B. The crisps advert. An advert says: *“9 out of 10 children prefer CrunchyChips!”* In small print: *“We asked 10 children at the CrunchyChips birthday party.”*

4. What percentage is 9 out of 10?
5. Give two reasons why you might not believe the advert.

C. The average age. Five children who are all 10 years old go on a trip with their 40-year-old teacher.

6. What is the average age of the group?
7. Someone says “the average age of the group is 15, so they’re teenagers”. What’s wrong with that?

D. The pictogram. A bike shop sold twice as many bikes this year. In its advert, it draws this year’s bike twice as tall *and* twice as wide as last year’s bike.

8. About how many times as big does the new picture look? (Hint: try drawing it on squared paper: a 1 × 1 square, then a 2 × 2 square.)

E. Finish.

9. In an honest bar chart that starts at 0, which bar is taller: 100 or 96?

   Talk about it: Which trick do you think works best on grown-ups? Make up your own sneaky graph and test it on someone at home.

Puzzle corner (optional)

Average detective.

- Write down five numbers that have an average of 6, where none of the numbers is 6.
- Can you find five numbers with an average of 6 where *all* the numbers are different and below 10?
- Can the average ever be bigger than all the numbers? Smaller than all of them? Try to find an example, or explain why not.

- Your average for three tests is 7. What mark do you need on the fourth test to raise your average to 7,5?